



Upbeat Nystagmus with an Unusual Velocity-Decreasing and Increasing Waveform: a Sign of Gaze-Holding Dysfunction in the Paramedian Tracts in the Medulla?

Jing Tian¹ · Jorge Otero-Millan^{1,2} · David S. Zee^{1,3,4,5} · Amir Kheradmand^{1,3,5,6} 

Accepted: 29 January 2022 / Published online: 8 February 2022

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract

We report a patient with spontaneous upbeat nystagmus (UBN) due to an ischemic lesion involving the paramedian tract (PMT) in the medulla. Eye movement recordings, using an infrared video-oculography (VOG) system, showed that the slow phase of the nystagmus was initially velocity-decreasing but gradually became velocity-increasing. Simulation of the nystagmus with a mathematical model supports a role for the PMT in relaying premotor signals for vertical gaze holding to the cerebellum. Our model shows that the disruption in cerebellar input from PMT can lead to the velocity-increasing waveform of the nystagmus, whereas the velocity-decreasing waveform could be related to a mismatch between the innervational commands to the ocular muscles (the pulse and step) needed to hold gaze steady.

Keywords Upbeat nystagmus · Paramedian tracts · Medulla · Waveform · Model · Gaze holding

Introduction

A neural network within the brainstem integrates eye velocity into position signals in order to maintain a steady gaze against the elastic restoring forces of the orbit [1, 2]. The neural integrators for gaze holding are comprised of circuits in the brainstem and the cerebellum in which projections from the cell groups of the paramedian tract (PMT) to the cerebellar flocculus and paraflocculus (tonsil) are thought to

be important [1, 3]. Little is known, however, about the exact mechanism and clinical relevance of lesions that affect this tract. Here, we report a patient with spontaneous upbeat nystagmus (UBN) due to an ischemic lesion involving the PMT in the rostral medulla. Each slow phase of the nystagmus exhibited a rapid velocity-decreasing drift initially, which was followed by a velocity-increasing drift. To understand how these abnormal patterns of nystagmus might emerge from mechanisms related to vertical gaze holding involving the PMT, we implemented a mathematical model to simulate the mixed waveforms of the nystagmus. Our model suggests that a disruption in the projection from the PMT to the cerebellum can lead to the velocity-increasing waveform of the nystagmus, whereas the velocity-decreasing waveform could be related to poor gaze holding due to a pulse-step mismatch.

Case Report

A 35-year-old woman with diabetes mellitus, hypertension, and HIV presented acutely with impaired vision due to vertical jumping of images (oscillopsia). She also had weakness and sensory changes on the left arm and leg. Ocular motor evaluation showed a spontaneous upbeat nystagmus (UBN). The brain MRI revealed an acute infarct on the right side of

✉ Amir Kheradmand
akherad@jhu.edu

¹ Department of Neurology, Johns Hopkins University School of Medicine, Baltimore, MD, USA

² School of Optometry, University of California, Berkeley, CA, USA

³ Department of Otolaryngology-Head and Neck Surgery, Johns Hopkins University School of Medicine, Baltimore, MD, USA

⁴ Department of Ophthalmology, Johns Hopkins University School of Medicine, Baltimore, MD, USA

⁵ Department of Neuroscience, Johns Hopkins University School of Medicine, Baltimore, MD, USA

⁶ Vestibular and Ocular Motor (VOR) Lab, Department of Neurology, The Johns Hopkins Hospital, Path 2-210, 600 N. Wolfe Street, Baltimore, MD 21287, USA

rostral medial medulla which included the paramedian tract (PMT) [4] (Fig. 1).

We used an infrared video-oculography (VOG) system (RealEyes xDVR, Micromedical Inc.) to record eye movements at a frame rate of 100 Hz. The eye movement recording was made at the bedside with the patient seated, looking at a handheld stationary target. The effect of removal of visual fixation on the spontaneous nystagmus was also assessed by recording in the dark. Eye movements were analyzed using the custom pupil tracking software [5]. A behavioral calibration of eye movements was unreliable in the presence of nystagmus and, instead, we assumed average values of 24 mm for the diameter of the globe and 12 mm for the diameter of the iris. Thus, a displacement of the center of the pupil that was equal to the radius of the iris corresponded to a movement of 30° on the VOG image.

Recordings showed an UBN with a variable slow-phase waveform (Fig. 2). Following a quick phase, the slow phase began with a rapid drift that had an exponentially decreasing velocity, which in most cases was followed by an increasing-velocity drift before the next quick phase (Fig. 2a). The UBN had a mean amplitude of 2.2° and a mean slow-phase velocity of $9.1^\circ/\text{s}$ (range of $4.1\text{--}14.1^\circ/\text{s}$) in the straight-ahead gaze position. The average time constant for the exponential decay of slow phases was 46 ms (range of 21–73 ms). With removal of visual fixation in the dark, some of the slow phases lasted up to three times longer than the slow phases with visual fixation, and the increasing velocity waveform became more apparent as the quick phases were less frequent, and there was more time for the slow-phase waveform to evolve (Fig. 2b and the zoom-in view). In addition, there were intermittent brief oscillations in the velocity trace (frequency ~ 12 Hz with peak-to-peak velocity $\sim 10^\circ/\text{s}$), which were better seen during extended slow-phase waveforms (asterisks, Fig. 2b). The nystagmus was attenuated temporarily after voluntary or spontaneous blinks (arrows, Fig. 2c). The waveform of the nystagmus did not change in different horizontal and vertical gaze positions, but there were

longer slow-phase waveforms following horizontal saccades (arrows, Fig. 2d). The patient had normal convergence, during which the nystagmus was attenuated compared to the straight-ahead gaze position (arrow, Fig. 2e).

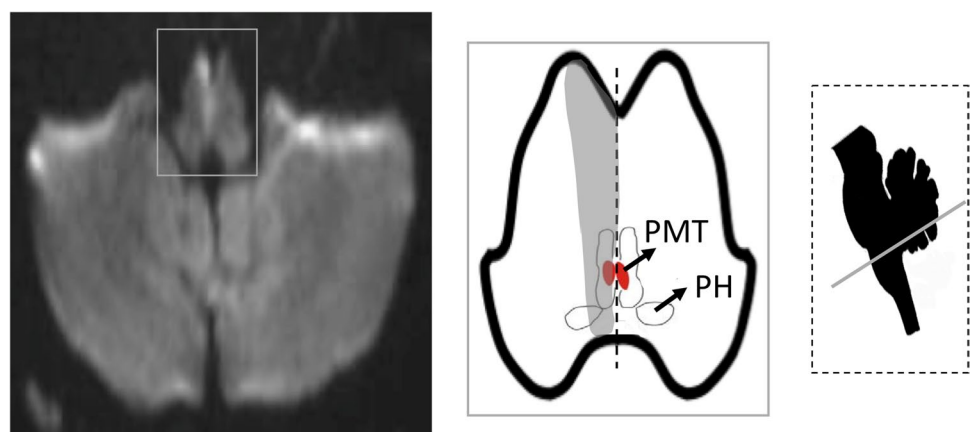
Discussion

Waveforms of Upbeat Nystagmus and Localized Lesions

Here, our patient with a medullary lesion involving PMT showed an UBN with mixed-velocity waveforms. The nystagmus was characterized by initially velocity-decreasing slow phases which then gradually became velocity-increasing. To our knowledge, UBN with such a mixed-velocity waveform has not been reported previously with brainstem lesions. A similar waveform has been described in a case of downbeat nystagmus in a patient with ankylosing spondylitis [6].

Vertical nystagmus (upbeat or downbeat) is commonly reported with lesions in the brainstem and cerebellar network that hold vertical gaze steady [1, 7, 8]. Different waveforms of nystagmus have been described with lesions at various locations. UBN with a linear slow phase has been reported with focal paramedian lesions in the caudal medulla [9, 10]. A constant-velocity UBN with a large amplitude (10°) has been also reported with a unilateral paramedian lesion in the upper pons [11], a unilateral lesion of the caudal pons [12], and a paramedian dorsal lesion of the caudal medulla [13]. UBN with an exponentially velocity-decreasing slow phase has been reported with a small infarct in the caudal dorsolateral medulla [14], unilateral medial medullary lesion [15], and with dorsomedial pontine lesion associated with bilateral internuclear ophthalmoplegia [16]. UBN with an increasing-velocity slow-phase waveform has been described in patients with large posterior fossa lesions [1, 17].

Fig. 1 Left: an axial cross section of the diffusion-weighted brain MRI shows an acute infarct in the right paramedian medulla which extends along the dorso-ventral axis. Middle: the approximate location of PMT (in red) within the infarct area and the relative location of the rostral perihypoglossal nuclei (PH) are illustrated on the axial section [4]. Right: the corresponding level of the infarct area is shown on the sagittal schematic of the brainstem



PMT and Gaze Holding

The gaze-holding network is composed of neural integrator circuits within the brainstem and cerebellum. In the brainstem, the neural integrator for vertical gaze holding primarily resides in the interstitial nucleus of Cajal (INC) within the midbrain [18] though it has connections with the structures in the medulla. As part of these caudal structures, the perihypoglossal nuclei of intercalatus and Roller relay vertical eye position signals to the cerebellum [1], and lesions in these nuclei may lead to upbeat nystagmus [19]. Within the cerebellum, the projections from the flocculus and paraflocculus improve the performance of inherently leaky brainstem neural integrators. In this context, the exponential decay in the slow phase of the nystagmus is suggestive of an impaired velocity-to-position integration, whereas the velocity-increasing slow phase can be caused by an instability in the cerebellar control of gaze holding [1].

The cell groups of PMT consist of several cell clusters located along the midline of the pons and medulla. There are at least six separate PMT subgroups which might be involved in vertical and horizontal gaze holding. One subgroup of PMT cells in the medulla receives vertical eye position signals from the INC and contributes to vertical gaze integration. Thus, lesions involving PMT or its connections with INC may cause a deficit in vertical gaze holding, and result in a nystagmus by disrupting the premotor gaze information to the cerebellum. Based on the afferent and efferent connections of PMT cell groups, it has long been suggested that PMT neurons could provide an efference copy of motor command to the floccular region [3, 20], and a simple PMT pathway has been used in computational models to relay the efference copy from the brainstem to the flocculus [21, 22]. Such a mechanism is in line with the gaze-evoked nystagmus found with PMT lesions [23]. While the precise mechanism of UBN is unknown in our patient and could be related to involvement of other structures within the brainstem such as the nucleus intercalatus of the perihypoglossal nuclei [24], the waveform of UBN could be consistent with abnormalities in the contribution of PMT to vertical gaze holding.

The neural integrator network also maintains gaze position with each type of eye movements. For example, a saccadic pulse signal must be integrated into a step signal to maintain gaze position following a saccade. Therefore, a gaze-holding dysfunction might be modulated by ocular motor inputs related to various types of eye movements. In our patient, likewise, saccades, convergence, and blinks could briefly suppress

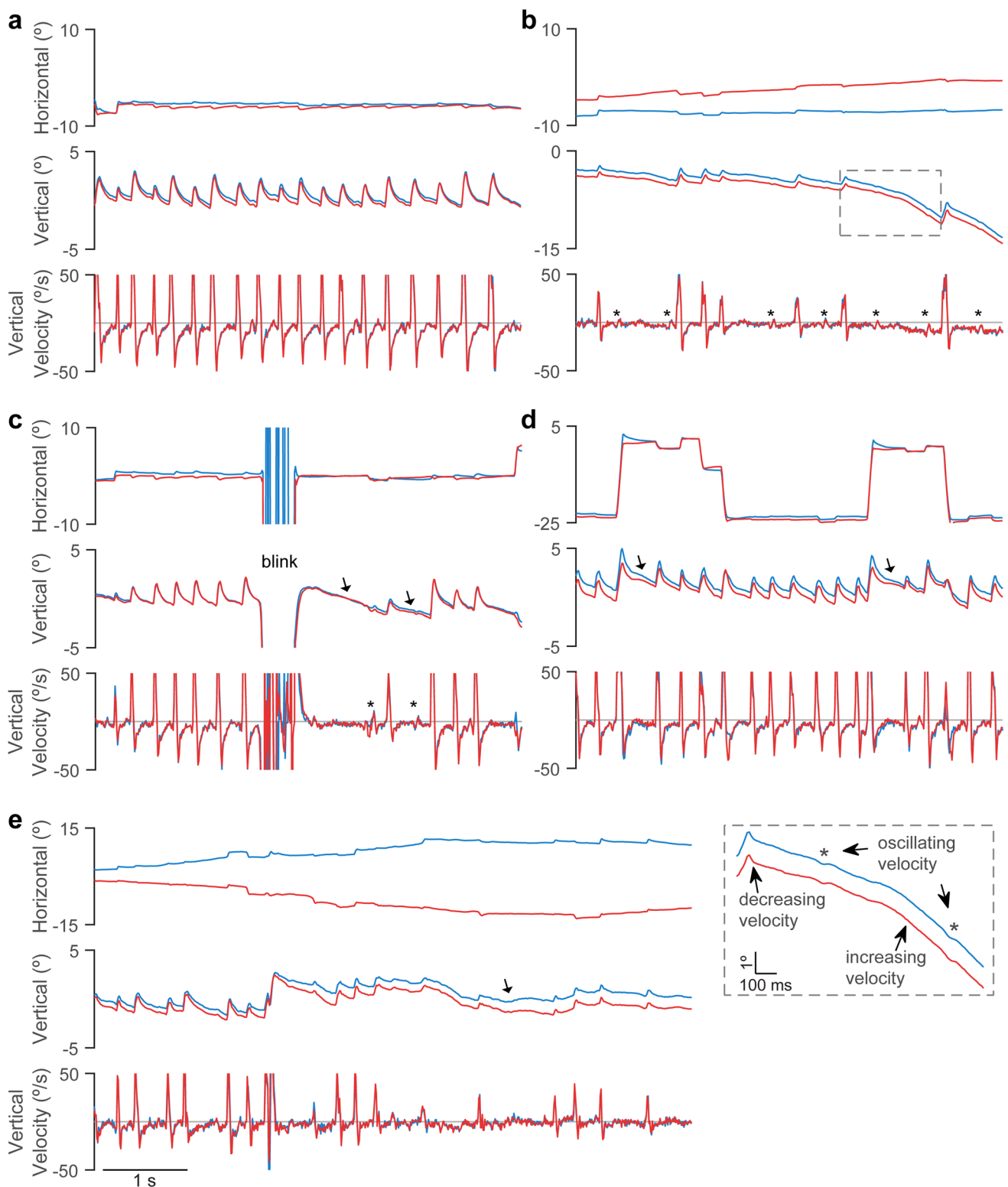
the quick phase of the nystagmus, and thus extend the duration of the slow phase and make it possible to see gradual changes in the waveform or other superimposed abnormalities such as the intermittent microflutters (Fig. 2).

Gaze Holding Model

To test our hypothesis that the nystagmus in our patient was caused by PMT dysfunction, we developed a mathematical simulation of the nystagmus (Fig. 3). This simulation is based on a model for vertical gaze holding that accounts for both cerebellar and brainstem pathways [21]. This model also contains the elements necessary to generate saccades and gaze holding. A bias signal that drives the vertical slow eye drift is sent directly from the vestibular nucleus (VN) to the ocular motor nuclei (OMN) and via the brainstem integrator. The premotor signals from the brainstem integrator are also relayed to the cerebellum (i.e., the flocculus) via PMT. The cerebellum contains a forward model that estimates the current eye velocity, which is then used to compute an error signal that encodes a desired eye velocity (i.e., zero in the case of gaze holding). This error signal is then amplified with a gain of g and fed back to inhibit the VN floccular target neurons [25]. In this model, the quick phases are generated via a simple burst generator that is driven by the retinal error. The burst is sent directly to the ocular motor nuclei (OMN) with a weight of K . It is also integrated by a leaky brainstem integrator and sent to OMN as a step signal.

Model Simulations

We implemented two elements in our gaze-holding model to account for the mixed slow-phase waveform of UBN with the PMT lesion (Fig. 4): (1) a downward drift bias and (2) a weight (gain) of INC projections to the flocculus via PMT. To generate an UBN, we simulated the gaze-holding behavior with a downward drift bias from VN to OMN directly and via the brainstem integrator (Fig. 4a). In the normal state, the PMT-flocculus feedback loop augments the leaky brainstem integrator (e.g., the time constant of 2 s can be raised to 20 s with a gain of 10), and thus it can maintain stable gaze holding. With an intact cerebellar feedback loop, as expected, the slow-phase velocity of a nystagmus is largely determined by the ratio of the ‘pulse’ command that generates the quick phase, to the ‘step’ command that encodes the new eye position during gaze holding. Therefore, the initial, velocity-decreasing slow phase would depend on the degree of mismatch between the pulse and step



commands. Our simulation suggests that a PMT lesion can result in a velocity-increasing slow phase by decreasing the weight (c) of cerebellar inputs (Fig. 4b). Thus,

overall, the mixed-velocity waveform of the nystagmus could be characterized by (i) the pulse-step mismatch causing the initial velocity-decreasing waveform, and

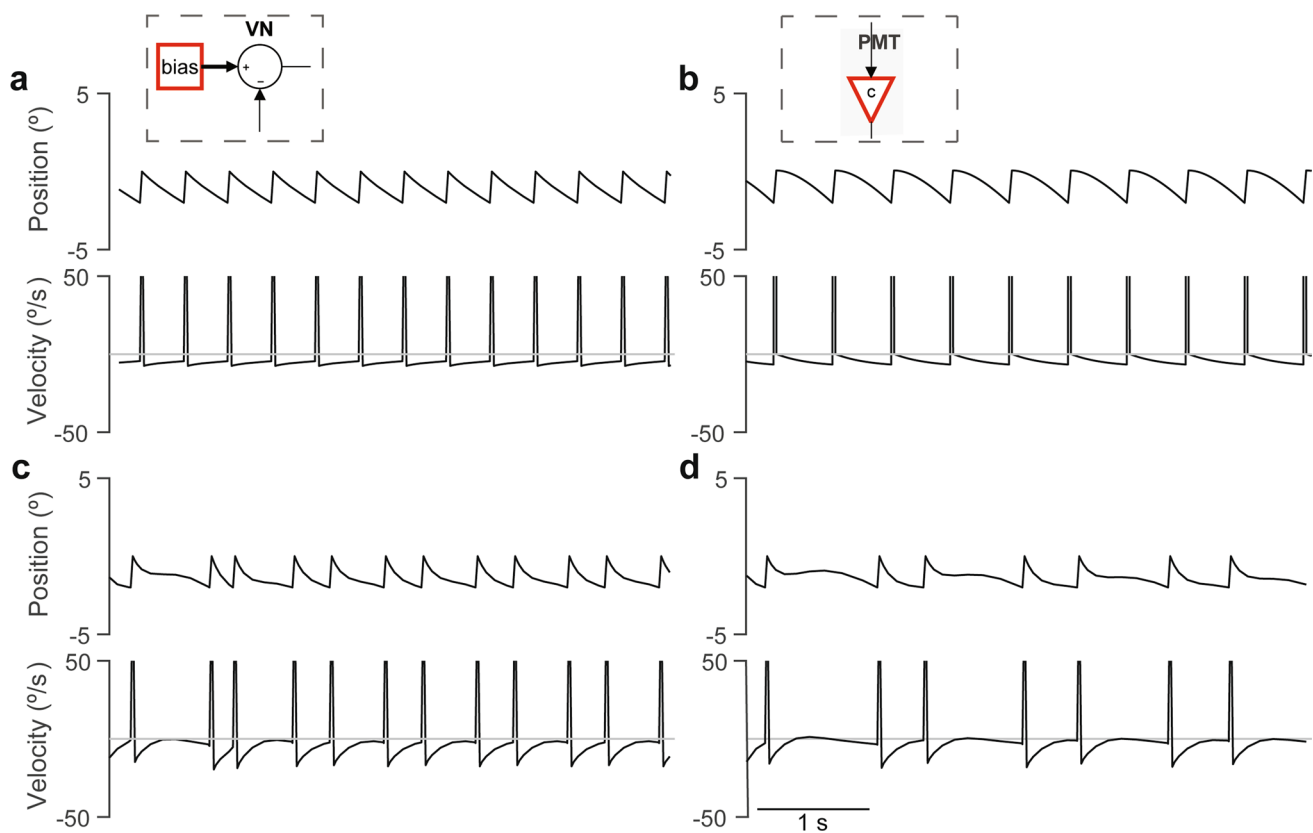


Fig. 4 Model simulations of UBN in the primary position of gaze (a–c) and in the dark (d). The corresponding element of the model (Fig. 3) is shown above each simulation (a, b). The brainstem integrator is modeled as leaky integrator with a time constant of 2 s. The negative velocity feedback raises the integrator time constant to about 20 s with a gain of 10. Simulated vertical eye position and velocity are plotted as a function of time. **a** Decreasing slow-phase velocity generated by a normal system (INC projection weight $c=1$)

with a downward drift bias (-0.8 rad/s) and by increasing the size of the pulse ($K=0.75$). **b** Increasing slow-phase velocity generated by partial damage to the PMT pathway from INC to the floccular region ($K=0.15$, $c=0.4$). **c** Mixed-velocity waveform generated by both pulse-step mismatch and partial damage to the PMT pathway ($K=0.75$, $c=0.4$). **d** Mixed-velocity waveform in the dark generated by reducing the bias (-0.5 rad/s) in simulation (c)

Author Contribution Jing Tian analyzed and interpreted the data, drafted the manuscript, and revised the manuscript.

Jorge Otero-Millan collected, analyzed, and interpreted the data and revised the manuscript.

David S. Zee interpreted the data and revised the manuscript.

Amir Kheradmand collected and interpreted the data, drafted the manuscript, and revised the manuscript.

Funding This work was supported by a grant from the Leon Levy foundation.

Data Availability All available data can be obtained by contacting the corresponding author.

Declarations

Ethics Approval The study was performed in accordance with the ethical standards laid down in the Declaration of Helsinki (1964). Approval was granted by the Johns Hopkins Institutional Review Board.

Consent to Participate The informed and written consent for publication of the data and videos was obtained from the patient.

Conflict of Interest The authors declare no competing interests.

References

1. Leigh RJ, Zee DS. The neurology of eye movements [Internet]. Oxford University Press; 2015 [cited 2018 Jul 26]. Available from: <http://www.oxfordmedicine.com/view/10.1093/med/9780199969289.001.0001/med-9780199969289>.
2. Shadmehr R. Distinct neural circuits for control of movement vs. holding still. *J Neurophysiol.* 2017; <https://doi.org/10.1152/jn.00840.2016>.
3. Büttner-Ennever JA, Horn AKE. Pathways from cell groups of the paramedian tracts to the floccular region. *Ann N Y Acad Sci.* 1996;781:532–40.
4. Buresch N. Neuroanatomische Charakterisierung blickstabilisierender Neurone an der Hirnstammmitellinie der Primaten, einschliesslich des Menschen [Dissertation]. Ludwig-Maximilians-Universität zu München; 2005.

5. Otero-Millan J, Roberts DC, Lasker A, Zee DS, Kheradmand A. Knowing what the brain is seeing in three dimensions: a novel, noninvasive, sensitive, accurate, and low-noise technique for measuring ocular torsion. *J Vis.* 2015;15:11.
6. Abel LA, Traccis S, Dell'Osso LF, Ansevin CF. Variable waveforms in downbeat nystagmus imply short-term gain changes. *Ann Neurol.* 1983;13:616–20.
7. Kim JS, Yoon B, Choi K-D, Oh S-Y, Park S-H, Kim B-K. Upbeat nystagmus: clinicoanatomical correlations in 15 patients. *J Clin Neurol Seoul Korea.* 2006;2:58–65.
8. Crnošija L, Adamec I, Klepac N, Borovečki F, Ozretić D, Habek M. Upbeat nystagmus due to medullary lesion: case report and literature review. *Medicine.* 2014;63:4.
9. Janssen JC, Larner AJ, Morris H, Bronstein AM, Farmer SF. Upbeat nystagmus: clinicoanatomical correlation. *J Neurol Neurosurg Psychiatry.* 1998;65:380–1.
10. Tilikete C, Hermier M, Pelisson D, Vighetto A. Saccadic lateropulsion and upbeat nystagmus: disorders of caudal medulla. *Ann Neurol.* 2002;52:658–62.
11. Pierrot-Deseilligny C, Milea D, Sirmay J, Papeix C, Rivaud-Péchox S. Upbeat nystagmus due to a small pontine lesion: evidence for the existence of a crossing ventral tegmental tract. *Eur Neurol.* 2005;54:186–90.
12. Tilikete C, Milea D, Pierrot-deseilligny C. Upbeat nystagmus from a demyelinating lesion in the caudal pons. *J Neuroophthalmol.* 2008;28:202–6.
13. Kim H-A, Yi H-A, Lee H. Can upbeat nystagmus increase in downward, but not upward, gaze? *J Clin Neurosci.* 2012;19:600–1.
14. Munro NA, Gaymard B, Rivaud S, Majdalani A, Pierrot-Deseilligny C. Upbeat nystagmus in a patient with a small medullary infarct. *J Neurol Neurosurg Psychiatry.* 1993;56:1126–8.
15. Hirose G, Ogasawara T, Shirakawa T, Kawada J, Kataoka S, Yoshioka A, et al. Primary position upbeat nystagmus due to unilateral medial medullary infarction. *Ann Neurol.* 1998;43:403–6.
16. Choi J-H, Jung N-Y, Kim M-J, Choi K-D, Kim JS, Jung D-S. Pure upbeat nystagmus in association with bilateral internuclear ophthalmoplegia. *J Neurol Sci.* 2012;317:148–50.
17. Traccis S, Rosati G, Aiello I, Monaco MF, Loffredo P, Puliga MV, et al. Upbeat nystagmus as an early sign of cerebellar astrocytoma. *J Neurol.* 1989;236:359–60.
18. Crawford JD, Cadera W, Vilis T. Generation of torsional and vertical eye position signals by the interstitial nucleus of Cajal. *Science.* 1991;252:1551–3.
19. Meling TR, Nouri A, May A, Guinand N, Vargas MI, Destrieux C. Upbeat vertical nystagmus after brain stem cavernoma resection: a rare case of nucleus intercalatus/nucleus of roller injury. *J Neurol.* 2020;267:2865–70.
20. Büttner-Ennever JA, Horn AK, Schmidtke K. Cell groups of the medial longitudinal fasciculus and paramedian tracts. *Rev Neurol (Paris).* 1989;145:533–9.
21. Marti S, Straumann D, Büttner U, Glasauer S. A model-based theory on the origin of downbeat nystagmus. *Exp Brain Res.* 2008;188:613–31.
22. Khojasteh E, Bockisch CJ, Straumann D, Hegemann SCA. A dynamic model for eye-position-dependence of spontaneous nystagmus in acute unilateral vestibular deficit (Alexander's Law). *Eur J Neurosci.* 2013;37:141–9.
23. Anagnostou E, Spengos K, Margeti S, Vassilopoulou S, Parakevas GP, Zis V. Vertical and horizontal integrator failure in a ponto-medullary infarction: a possible role for paramedian tract neurons. *J Neurol Sci.* 2009;280:118–9.
24. Saito T, Aizawa H, Sawada J, Katayama T, Hasebe N. Lesion of the nucleus intercalatus in primary position upbeat nystagmus. *Arch Neurol.* 2010;67:1403–4.
25. Zhang Y, Partsialis AM, Highstein SM. Properties of superior vestibular nucleus flocculus target neurons in the squirrel monkey. I. General properties in comparison with flocculus projecting neurons. *J Neurophysiol.* 1995;73:2261–78.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.